Examiner only

2. A 30 kW gas boiler is used for **700 hours** in 1 year.

(a) (i) Use the equation:

units used (kWh) = power (kW) × time (h)

to calculate the units used in 1 year. [2]

units used = ... kWh

(ii) Use the equation:

cost = units used × cost per unit

to calculate the cost of using the boiler for 700 hours. [2]

One unit of gas costs £0.12.

cost = £ ....

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(iii) Use your answer in (a)(ii) to calculate the cost of using the boiler for **1 hour**. [1]

cost = £ ....

(b) A solar water heating system costs £3600.

It is estimated to reduce gas consumption so that the boiler is used for only 650 hours instead of 700 hours.

(i) Calculate the saving made in 1 year by using the boiler for 50 hours less. [1]

saving = £ ....

(ii) Calculate the payback time of the cost of the solar water heating system. [1]

payback time = ... years

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